

# Proposal Presentation

Comparing Similarity Measures from Different  
Paradigms  
for Time Series Clustering

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# Motivation & Background

## Why This Study?

**Similarity measures** are the core component of time series clustering. Choosing the right one directly determines clustering quality.

Existing benchmarks cover traditional measures (ED, DTW, SBD, GAK) do NOT include distributional kernel (IDK)

No study has systematically compared IDK against traditional measures under a unified clustering framework.

Few studies analyze how data characteristics (noise, shift, length) affect measure performance.

# Motivation & Background

## The Gap

Existing large-scale benchmarks (*Paparrizos VLDB'25: 128 datasets*) cover lock-step, elastic, sliding, and some kernel measures — but **not IDK**.

**CTDS (Gong et al., PAKDD 2024)** uses IDK for time series clustering, but its goal is to propose a new method — not a systematic comparison of measures.

# Aim

To compare different types of **similarity measures** for time series clustering and analyze how data characteristics (noise, temporal shift, and length) affect their performance.



lock-step (Euclidean Distance), elastic (DTW, MSM), sliding (SBD), and distributional kernel (IDK)

# Research Questions

## RQ1

How does IDK (distributional paradigm) compare against traditional measures (ED, DTW, MSM, SBD) for clustering?

## RQ2

How does increasing noise affect the relative performance of each measure?

## RQ3

How does temporal misalignment (shift) impact different similarity measures?

## RQ4

How do differences in sequence length affect similarity computation and clustering quality?

# Relevant Papers

## BENCHMARK

### Paparrizos & Bogireddy (VLDB 2025)

Most comprehensive TS clustering benchmark. 128 UCR datasets, covers ED/DTW/MSM/SBD/GAK/SINK. IDK not included.

*Our difference: We extend by adding IDK to the comparison.*

## IDK THEORY

### Ting et al. (VLDB Journal 2024)

Proposes R-domain approach: treating time series as distributions. Uses IDK for anomaly detection.

*Our difference: We apply the distributional paradigm to clustering.*

## IDK CLUSTER

### Gong et al. (PAKDD 2024) — CTDS

Uses IDK for TS clustering via local subsequence distributions. Proposes a new method.

*Our difference: We do fair measure comparison, not propose new method.*

## DISTANCE

### Paparrizos & Liu (SIGMOD 2020)

Evaluates 52 distance measures with 8 normalizations on 128 datasets via 1-NN classification.

*Our difference: We focus on the unsupervised (clustering) setting.*

# Scope of the Study

## Included

Univariate whole time series clustering

5 similarity measures from 4 paradigms

15–20 datasets from UCR archive

Controlled perturbation analysis (noise, temporal shift, length variation)

Statistical significance testing (Friedman + Nemenyi post-hoc test)

## Excluded

Deep learning-based clustering methods

Multivariate time series

Feature-based / model-based representations

Supervised / semi-supervised settings

# Methodology

## Algorithm: K-medoids

| Paradigm       | Measure        | Key Property                           | Why Selected                                                      |
|----------------|----------------|----------------------------------------|-------------------------------------------------------------------|
| Lock-step      | Euclidean (ED) | Point-to-point alignment               | Baseline; most common; $O(n)$ complexity                          |
| Elastic        | DTW            | Time-axis warping (one-to-many)        | Most popular elastic measure; state-of-the-art for classification |
| Elastic        | MSM            | Edit-based (move-split-merge)          | Recent high-performer; metric property                            |
| Sliding        | SBD            | Cross-correlation (shift-invariant)    | Core of k-Shape; best clustering performance in VLDB'25 benchmark |
| Distributional | IDK            | Distribution-based; no point alignment | Novel R-domain paradigm; not yet benchmarked for clustering       |



# Methodology — Experiment Design

## Part 1: Real-Data Benchmark

### 15–20 UCR datasets

(diverse lengths, domains, # classes)

5 measures × k-medoids

Evaluate: ARI, NMI

Statistical test: Friedman + Nemenyi

## Part 2: Controlled Perturbation

### 3–5 representative datasets

Noise: Add Gaussian noise (varied SNR)

Shift: Random temporal displacement

Length: Truncate / pad sequences

→ Performance degradation curves

# Datasets & Evaluation

## Datasets

**Source:** UCR Time Series Archive

([timeseriesclassification.com](https://timeseriesclassification.com))

### Selection criteria:

- Diverse time series lengths (short → long)
- Varied number of classes (2 → 10+)
- Multiple application domains
- Equal-length series for fair comparison
- 15–20 datasets selected

## Evaluation Measures

**ARI**

Adjusted Rand Index  
Chance-corrected pairwise agreement

**NMI**

Normalized Mutual Information  
Information-theoretic measure

### Statistical Testing

Friedman test (global comparison)  
→ Nemenyi post-hoc test (pairwise)  
→ Critical Difference (CD) diagram  
Following standard practice in Paparrizos VLDB'25 benchmark.

# Software & Tools

Language

**Python**

IDK

**Isolation Kernel**

Official open-source implementation on GitHub

Statistics

**scipy / scikit-posthocs**

Friedman test & Nemenyi post-hoc test  
for statistical significance analysis

TS Library

**tslearn / aeon**

For traditional measures;  
k-medoids clustering with custom distance matrices

ML Library

**scikit-learn**

Evaluation metrics (ARI, NMI);  
Data preprocessing

Visualization

**matplotlib**

Critical Difference diagrams;  
performance degradation curves

# Planning & Work Split

| Week   | Chen                                             | Wang                                           |
|--------|--------------------------------------------------|------------------------------------------------|
| Week 1 | Literature review<br>+ dataset selection         | Set up experiment<br>framework + test IDK code |
| Week 2 | Implement ED / DTW / MSM<br>+ run Part 1         | Implement SBD / IDK<br>+ run Part 1            |
| Week 3 | Complete Part 1<br>+ start Part 2 perturbation   | Part 2 experiments<br>+ visualization          |
| Week 4 | Write Introduction<br>Related Work / Methodology | Write Results<br>Discussion / Conclusion       |
| Week 5 | Cross-review<br>+ integrate report               | Cross-review<br>+ final polish & submit        |

**Due: Friday 6 June 2026 (Week 13)**

# References

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# Questions & Discussion